

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

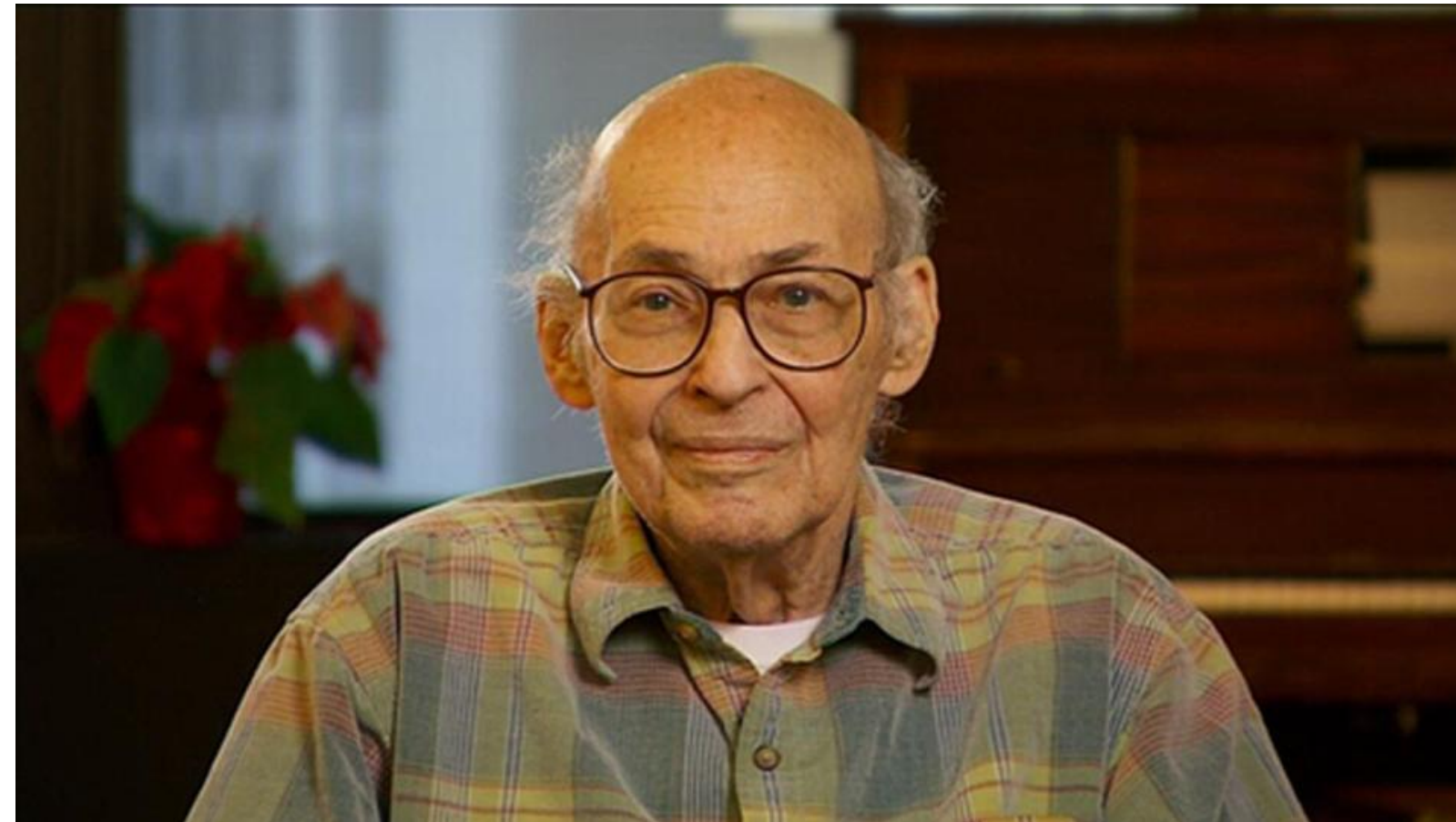
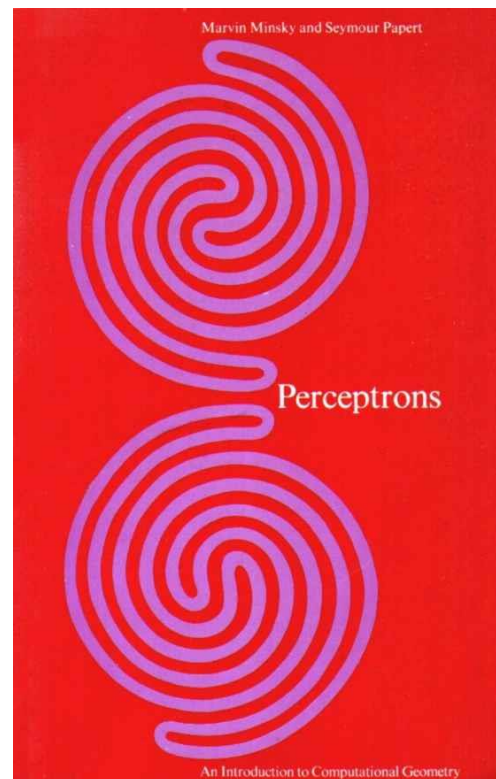
Deep Learning Models

Texas A&M University

Outline

- History of Deep Learning
- Convolutional Neural Networks
- Recurrent Neural Networks
- Transfer Learning
- Multi-task Learning

Deep Learning (i): History and Basics



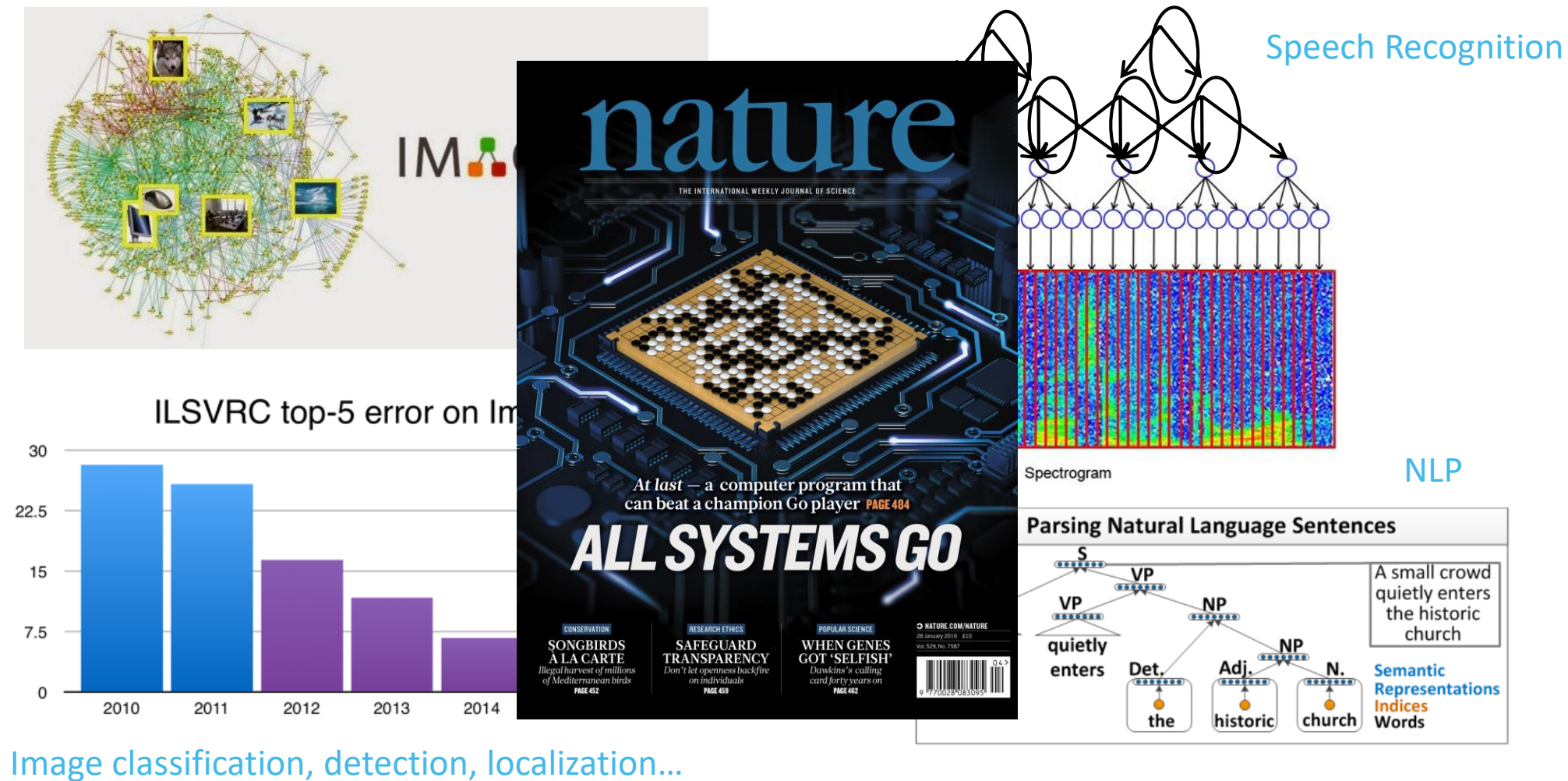
Marvin L. Minsky
Turing Awardee 1969, “Father of Artificial Intelligence”
(August 9, 1927 – January 24, 2016)

Who co-founded MIT’s CSAIL, MIT Media Lab, and “killed neural networks once” in 1970s

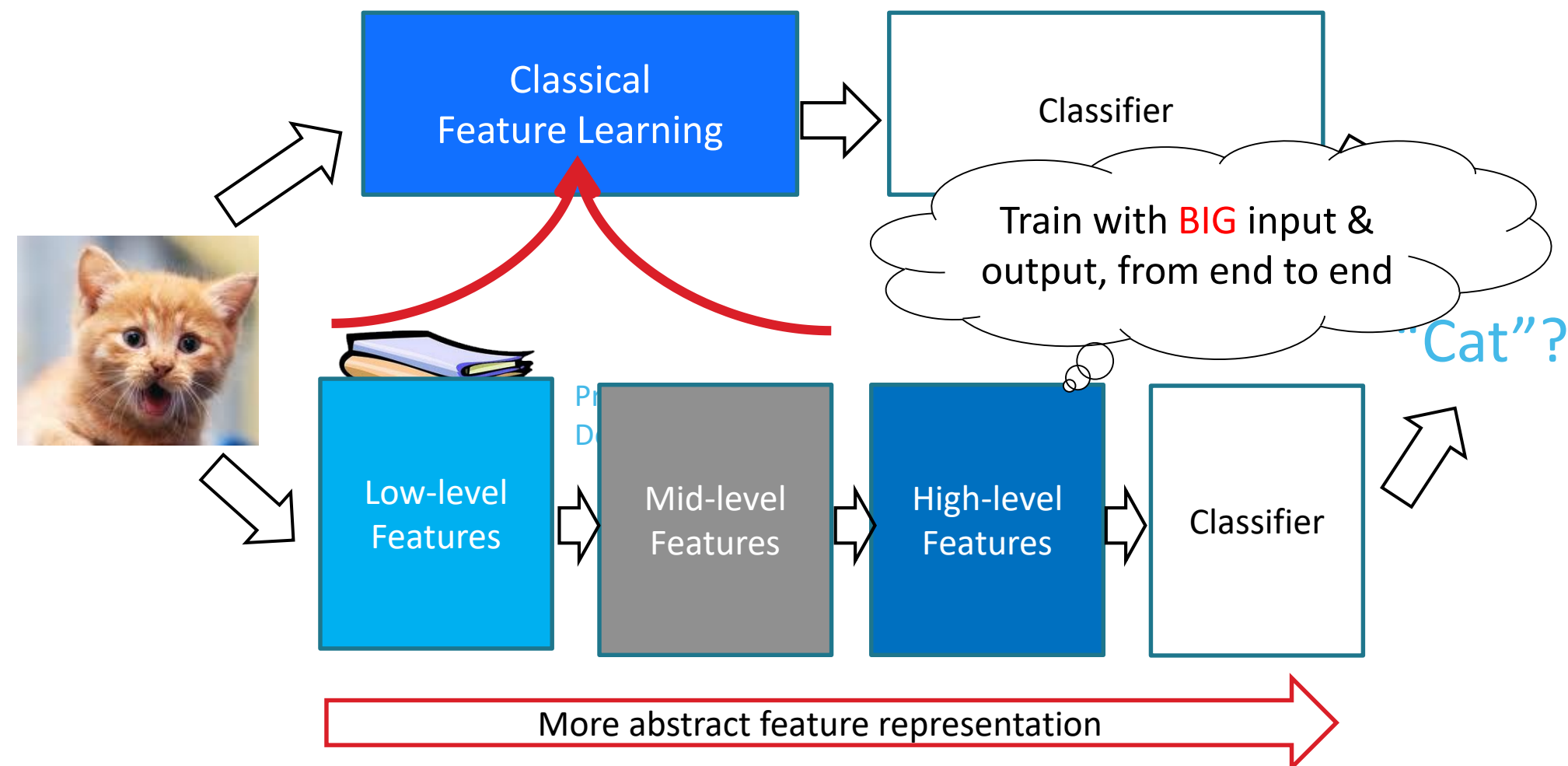


A Triumph of Deep Learning: 2012 - present

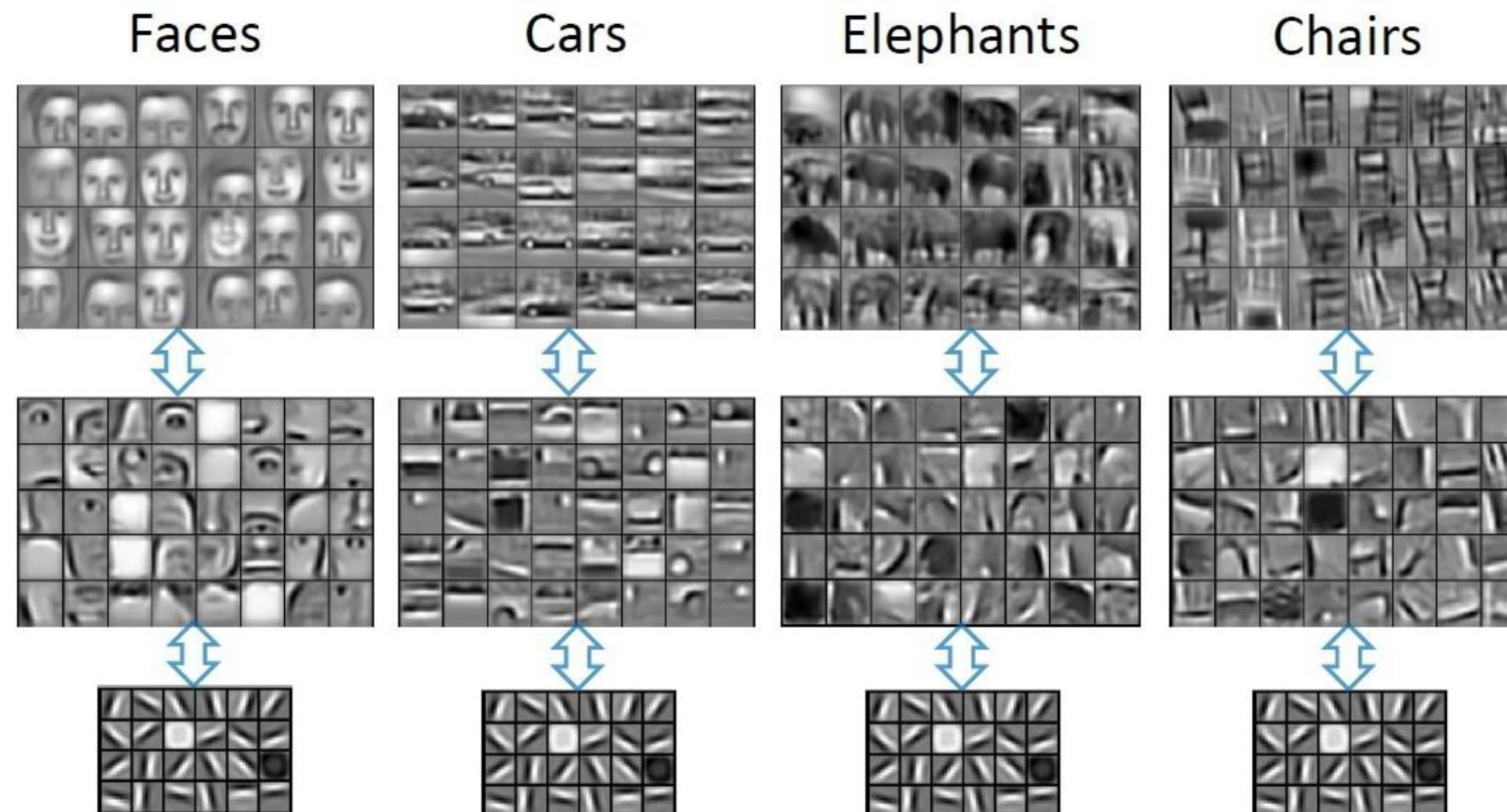
Top-performers in many tasks, over many domains



Feature learning: Going Deep



Deep Features (May) Learn Semantic Hierarchy



Status Quo

AlexNet, 8 layers
(ILSVRC 2012)



VGG, 19 layers
(ILSVRC 2014)



ResNet, 152 layers
(ILSVRC 2015)



Current Trend:

- To build increasingly larger, deeper networks, trained with more massive data, based on the benefits of high-performance computing.
- Play with the connectivity and add “skips”